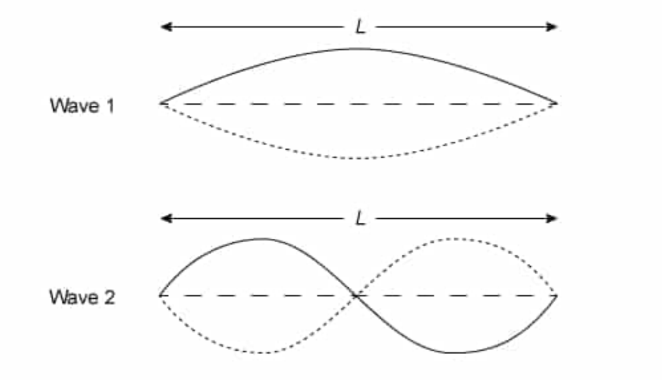
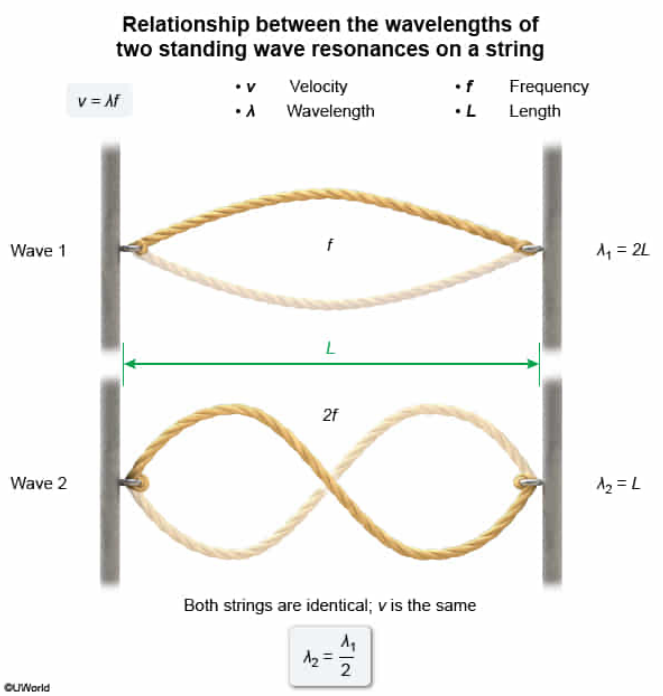


Two standing waves are created on two identical strings, as shown in the figure above. In wave 1, the standing wave vibrates at a frequency *f*, and in wave 2 the standing wave vibrates at a frequency 2*f*. How do the wavelengths λ_1 and λ_2 on each string compare?

- A. $\lambda_2 = \frac{\lambda_1}{4}$ [6%]
- B. $\lambda_2 = \frac{\lambda_1}{2}$ [75%]
- C. $\lambda_2 = \lambda_1$ [2%]
- D. $\lambda_2 = 2\lambda_1$ [16%]

Correct
75% answered correctly

Explanation:



A **wave** is a motion created by a disturbance in a medium. A wave moves through the medium with a speed *v* that is the product of the wavelength λ and the frequency *f* of the wave:

$$v = \lambda f$$

However, the **speed of waves** is determined only by the physical properties of the medium through which they travel.

In this question, two different standing waves are created on identical strings. Because the medium is the same, the velocities of the waves in each case are equal:

$$v_1 = v_2 = v$$

Wave 1 vibrates with frequency *f*, and wave 2 vibrates with frequency 2*f*. Hence, the wavelengths λ_1 and λ_2 on each string can be determined from equations for wave speed:

$$v = f\lambda_1 \quad v = (2f)\lambda_2$$
$$\lambda_1 = \frac{v}{f} \quad \lambda_2 = \frac{v}{2f}$$

Therefore, the relationship between the wavelengths is:

$$\lambda_2 = \frac{\lambda_1}{2}$$

(Choice A) The wavelengths differ by a factor of 2, not a factor of 4.

(Choice C) The wavelengths cannot be equal because although the wave speeds are the same, the frequency of each wave is different.

(Choice D) λ_2 is smaller than λ_1 because wave 2 has a greater frequency. λ_2 equals the length *L* of the string, and λ_1 equals 2*L*.

Educational objective:
When waves travel through the same medium, their speed is constant, and wavelength is inversely proportional to frequency.

Time spent:0 sec
Subject:
Physics

QID:403961
System:
Light and Sound